
Instructions. (15 points) Sections 12.6, 13.1–13.2. Show relevant work for full credit.

- (9pts) 1. Consider the surface $z = 4 - x^2 - 4y^2$.
- (a) (2 pts) Find the horizontal traces ($z = k$). For what values of k do real traces exist?
- (b) (2 pts) Find the vertical traces in the xz -plane ($y = 0$) and the yz -plane ($x = 0$).
- (c) (3 pts) Sketch the surface. Label the vertex and at least one trace in each direction.
- (d) (2 pts) Sketch a contour map (in the xy -plane) showing the level curves for $z = 0$, $z = 2$, and $z = 4$. Label each curve with its z -value.

(6^{pts}) **2.** Consider the helix $\mathbf{r}(t) = \langle 3 \cos t, 3 \sin t, 4t \rangle$.

(a) (2 pts) Find the first point (with $t > 0$) where the curve pierces the xz -plane.

(b) (2 pts) Find the unit tangent vector $\mathbf{T}$ at the point from (a).

(c) (2 pts) Evaluate $\int_0^{\pi/2} \mathbf{r}(t) dt$.